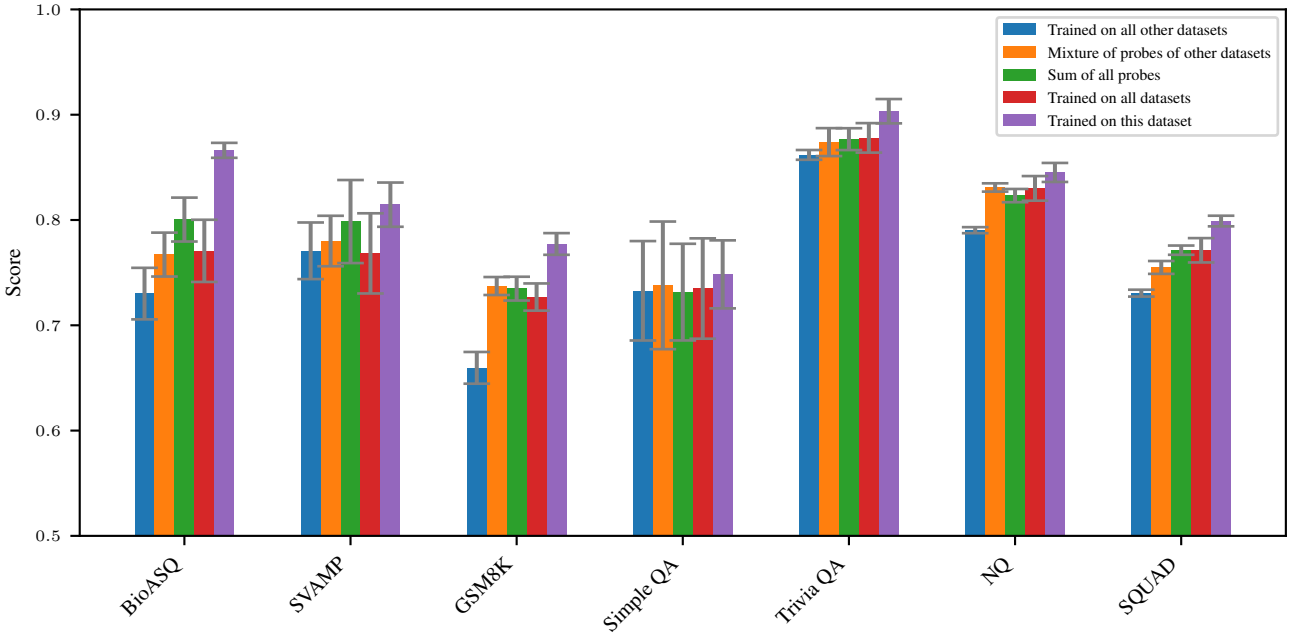

Figure 28. AUROC of linear probes at the stop token of the output in the multi-task setting, using L2 regularization.

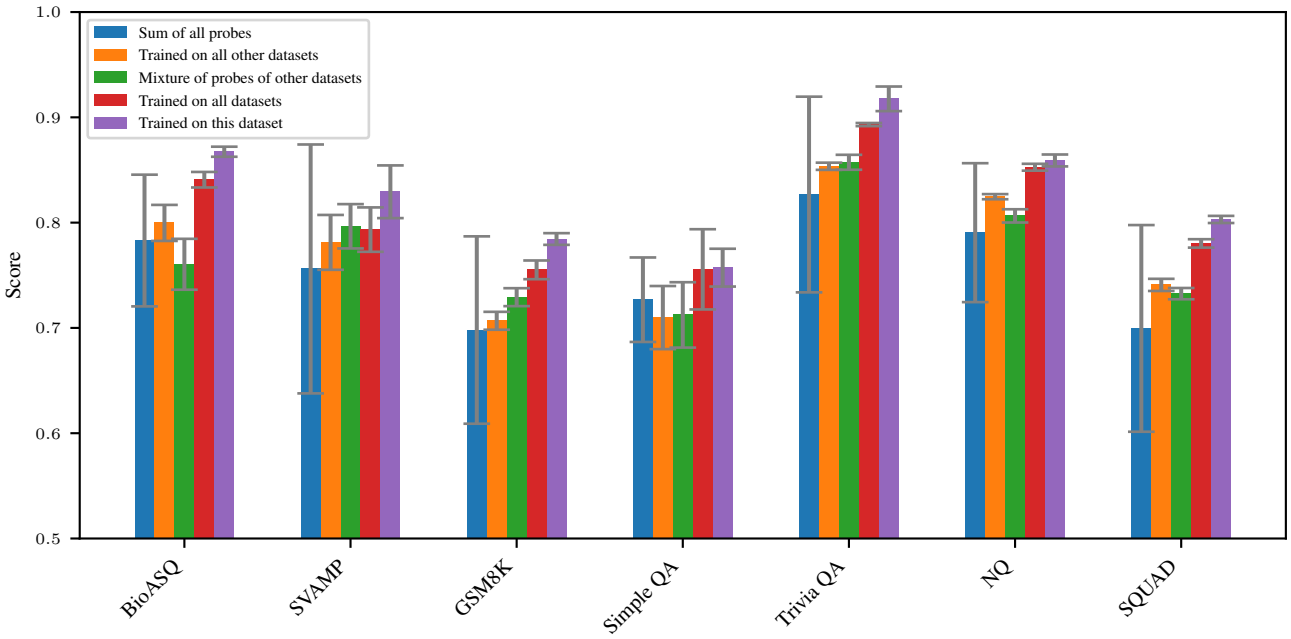

Figure 29. AUROC of linear probes at the token before the stop token of the output in the multi-task setting, using L2 regularization.

A.2. Layer 21

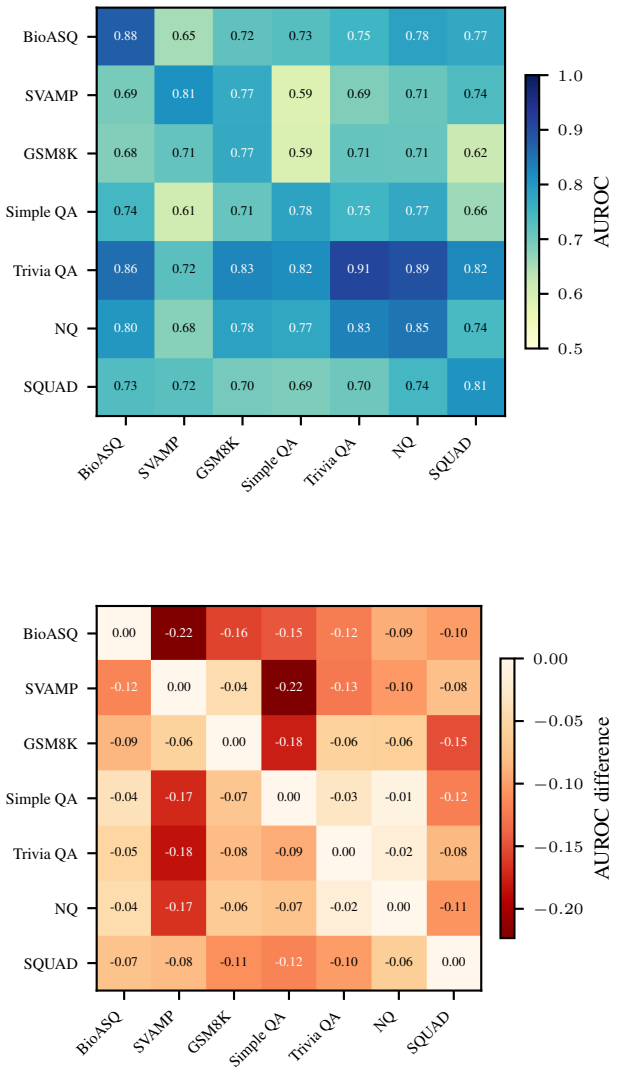

Figure 30. AUROC of probes trained on different tasks on the stop token of the output on last layer. Rows correspond to evaluation tasks while columns correspond to training tasks. The second plot represents the difference between the probe trained on this task and probes trained on the other datasets. Results are averaged over 5 runs.

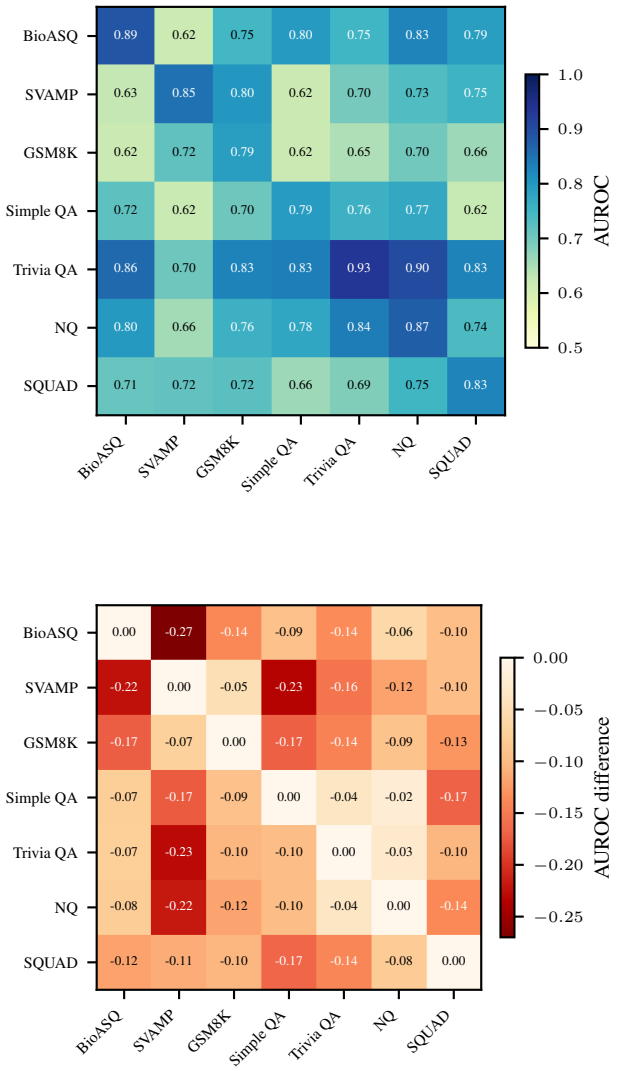

Figure 31. AUROC of probes trained on different tasks on the token before the stop token of the output on last layer. Rows correspond to evaluation tasks while columns correspond to training tasks. The second plot represents the difference between the probe trained on this task and probes trained on the other datasets. Results are averaged over 5 runs.

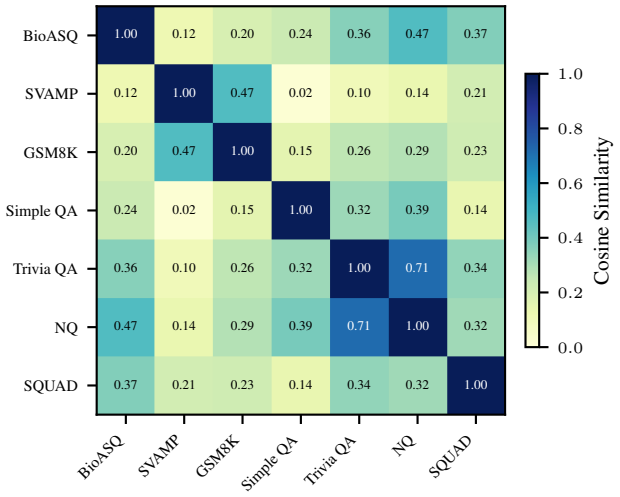

Figure 32. Cosine similarity between probes trained on different datasets using L2 regularization. Results are averaged over 5 runs.

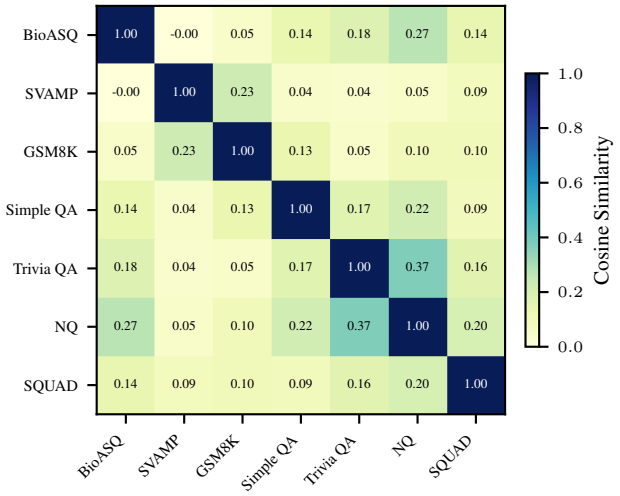

Figure 33. Cosine similarity between probes trained on different datasets using L2 regularization. Results are averaged over 5 runs.

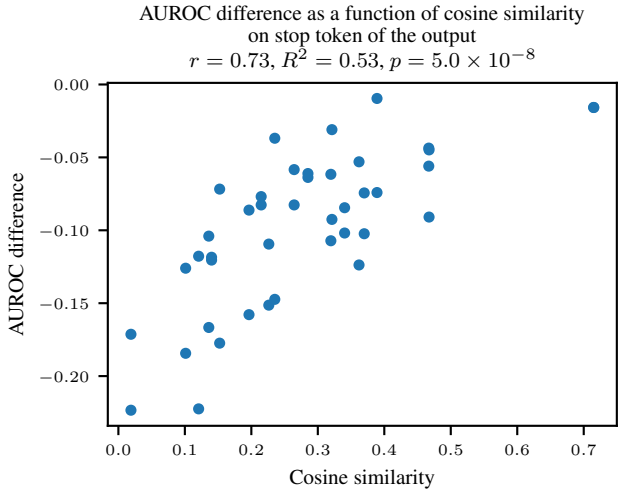

Figure 34. AUROC difference to probe trained on the right dataset as a function of cosine similarity between probes. Results are averaged over 5 runs.

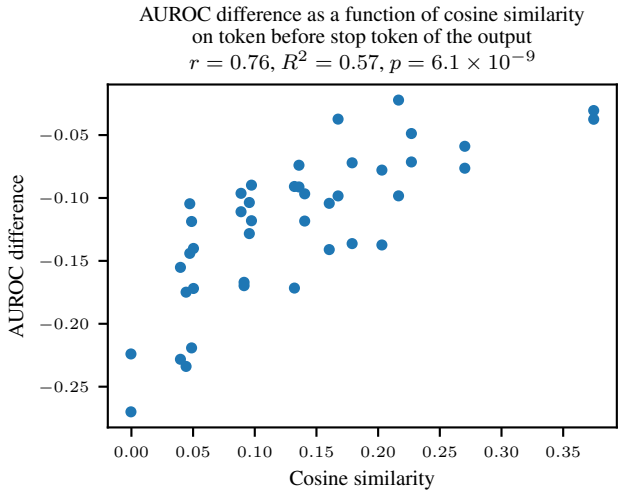

Figure 35. AUROC difference to probe trained on the right dataset as a function of cosine similarity between probes. Results are averaged over 5 runs.